



Course Specification

(Bachelor)

Course Title: Data Structures

Course Code: 501324-3

Program: Bachelor in Computer Science

Department: Department of Computer Science

College: College of Computers and Information Technology

Institution: Taif University

Version: 2

Last Revision Date: 22-02-2025

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Computer Science Department



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A. General information about the course:

1. Course Identification

1. Credit hours: (3)				
2. Course type				
A.	☐ University	☒ College	☐ Department	☐ Track
B.	☒ Required		☐ Elective	
3. Level/year at which this course is offered: (Level: 2, Year: 1)				
4. Course General Description:				
Introduces students to development, implementation, and analysis of efficient data structures and algorithms. Topics include analysis of sorting and searching algorithms and abstract data types such as stacks, queues, linked list, trees, graphs, and hash tables.				
5. Pre-requirements for this course (if any):				
Computer Programming (501121-4)				
6. Co-requisites for this course (if any):				
-				
7. Course Main Objective(s):				
1. To develop students' knowledge in data structures and the associated algorithms, introduce the concepts and techniques of structuring and operating on Abstract Data Types in problem solving, 2. To discuss common sorting, searching and graph algorithms, 3. To study the complexity and comparisons among these various techniques 4. To apply and implement learned algorithm design techniques and data structures to solve problems.				

2. Teaching mode (mark all that apply)

No	Mode of Instruction	Contact Hours	Percentage
1	Traditional classroom	3	100%
2	E-learning	-	-
3	Hybrid • Traditional classroom • E-learning	-	-
4	Distance learning	-	-

3. Contact Hours (based on the academic semester)





No	Activity	Contact Hours
1.	Lectures	45
2.	Laboratory/Studio	0
3.	Field	0
4.	Tutorial	0
5.	Others (specify)	0
Total		45

B. Course Learning Outcomes (CLOs), Teaching Strategies and Assessment Methods

Code	Course Learning Outcomes	Code of PLOs aligned with the program	Teaching Strategies	Assessment Methods
1.0	Knowledge and understanding			
1.1	Understand pointer and the linked list data structure to solve problems.	K1	Lecture Problem Solving	Direct: Exams Indirect: Course Exit Survey
1.2	Understand the stack and queues data structures to solve problems.	K1	Lecture Problem Solving	Direct: Exams Indirect: Course Exit Survey
...				
2.0	Skills			
2.1	Apply basic data structures in implementing practical applications using several data structures.	S1	Lectures Discussion	Direct: Quizzes, Homework, Exams Indirect: Course Exit Survey
2.2	Apply different data structures (binary tree, binary search tree, hash, graph etc.) to solve problems	S1	Lectures Discussion	Direct: Quizzes, Homework, Exams Indirect: Course Exit Survey
2.3	Analyze the time and space efficiency of algorithms.	S1	Lectures	Direct: Quizzes, Homework, Exams





Code	Course Learning Outcomes	Code of PLOs aligned with the program	Teaching Strategies	Assessment Methods
				Indirect: Course Exit Survey
3.0	Values, autonomy, and responsibility			
-	-	-	-	-

C. Course Content

No	List of Topics	Contact Hours
1.	Introduction to Data Structures and understanding standard data structures.	3
3.	Linked Lists	6
4.	Stacks and Queues	6
5.	Trees: Binary search tree, Binary Heap	9
6.	Hash Tables	3
7.	Basics of Algorithm Analysis	3
8.	Searching Algorithms	3
9.	Sorting Algorithms	6
10.	Graph data structures	6
Total		45

D. Students Assessment Activities

No	Assessment Activities *	Assessment timing (in week no)	Percentage of Total Assessment Score
1.	Homework /Quiz/Exercise/Class Participation	Week 1-15	40%
2.	Mid-Term	Week 7	20%
3.	Final Exam	Week 15	40%

*Assessment Activities (i.e., Written test, oral test, oral presentation, group project, essay, etc.).

E. Learning Resources and Facilities

1. References and Learning Resources

Essential References	Text Book: Data Structures and Algorithm Analysis in C++ by Mark Allen Weiss, 2013
Supportive References	Algorithms and Data Structures by N. Wirth Lectures Slides



Electronic Materials	Blackboard
Other Learning Materials	-

2. Required Facilities and equipment

Items	Resources
facilities (Classrooms, laboratories, exhibition rooms, simulation rooms, etc.)	A Lecture room appropriate for maximum 25 students with a personal computer, a data show and a smart board.
Technology equipment (projector, smart board, software)	Video projector / data show
Other equipment (depending on the nature of the specialty)	

F. Assessment of Course Quality

Assessment Areas/Issues	Assessor	Assessment Methods
Effectiveness of teaching	Students	Students' surveys and students' course evaluation
Effectiveness of students' assessment	Course Coordinator	Student evaluation, faculty input, course file, and program assessment
Quality of learning resources	Curriculum Committee	Review of CAF (Course assessment file) and student surveys
The extent to which CLOs have been achieved	Curriculum Committee	Evaluation and analysis of CLO and PLO assessment results
Other		

Assessors (Students, Faculty, Program Leaders, Peer Reviewers, Others (specify))

Assessment Methods (Direct, Indirect)

G. Specification Approval

COUNCIL /COMMITTEE	COMPUTER SCIENCE COUNCIL
REFERENCE NO.	MEETING #11
DATE	6 APRIL 2025